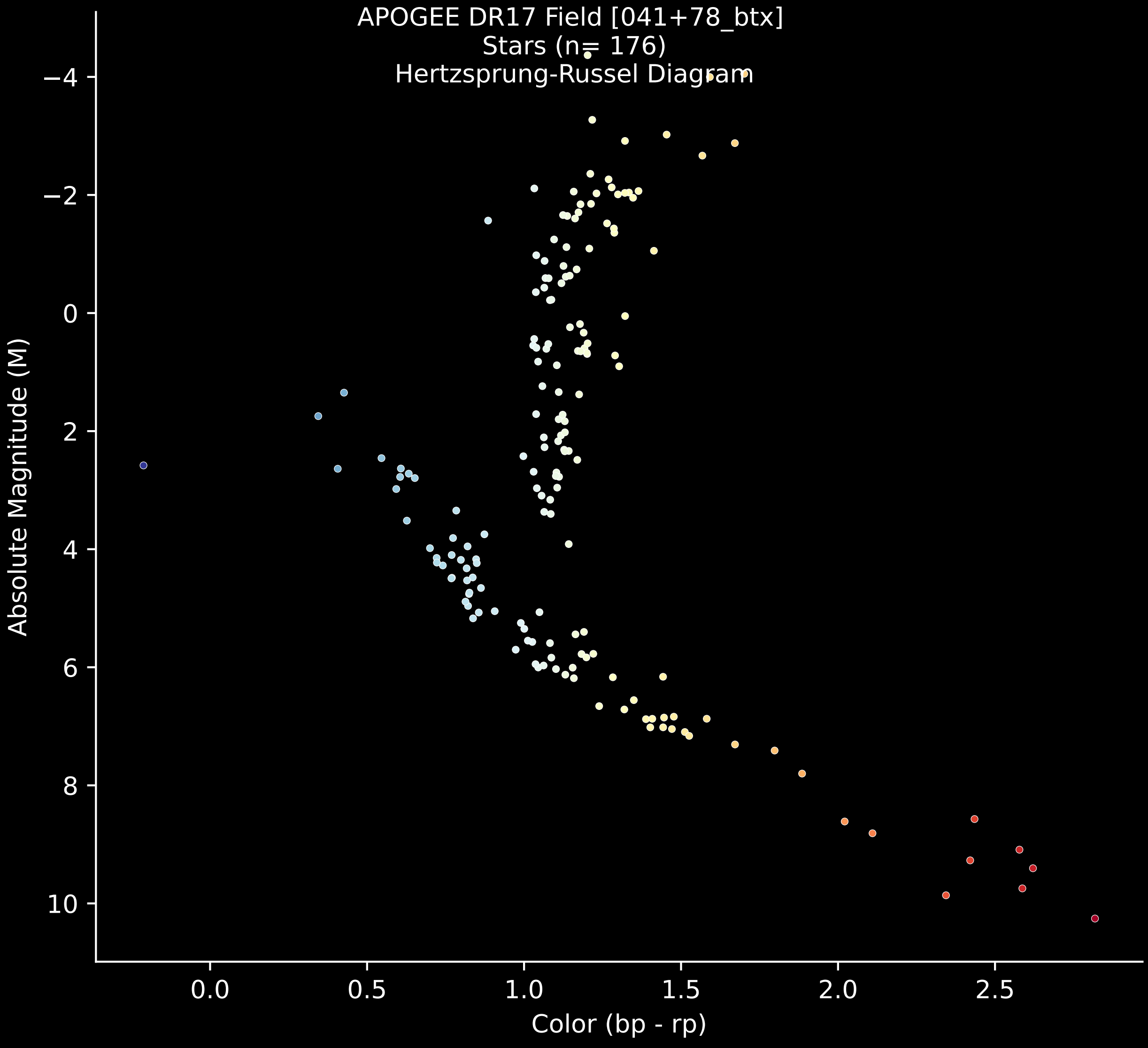
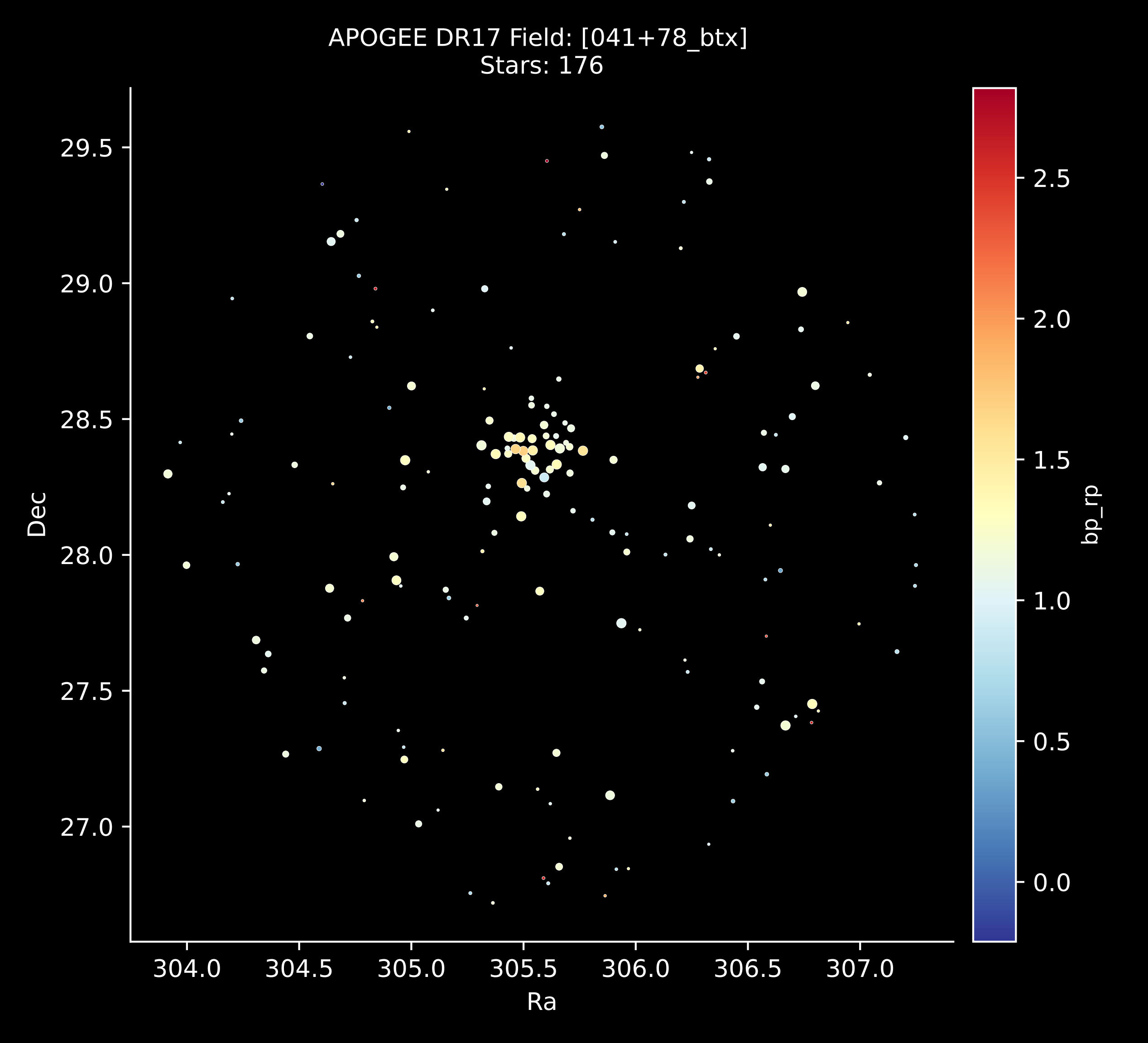
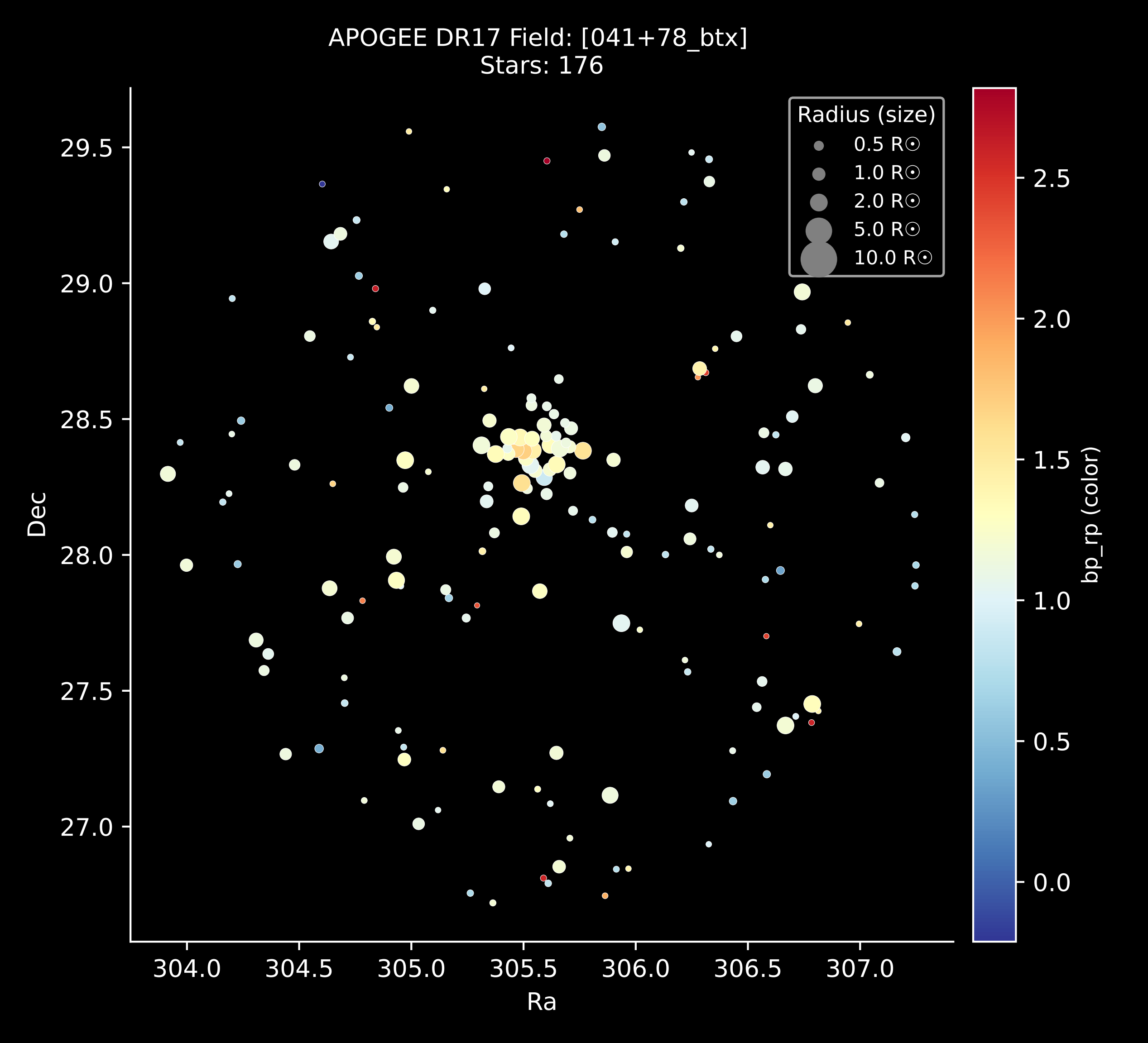
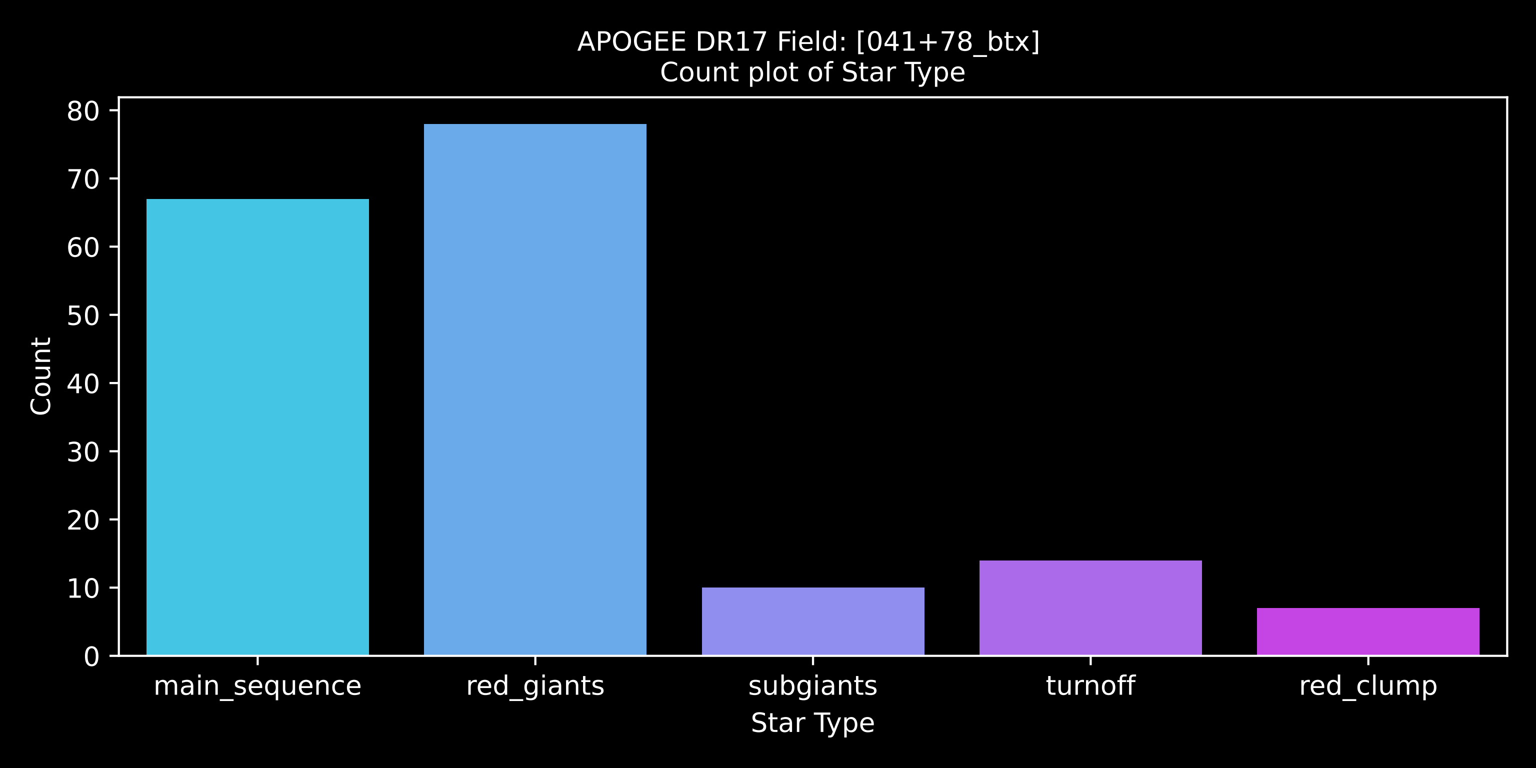
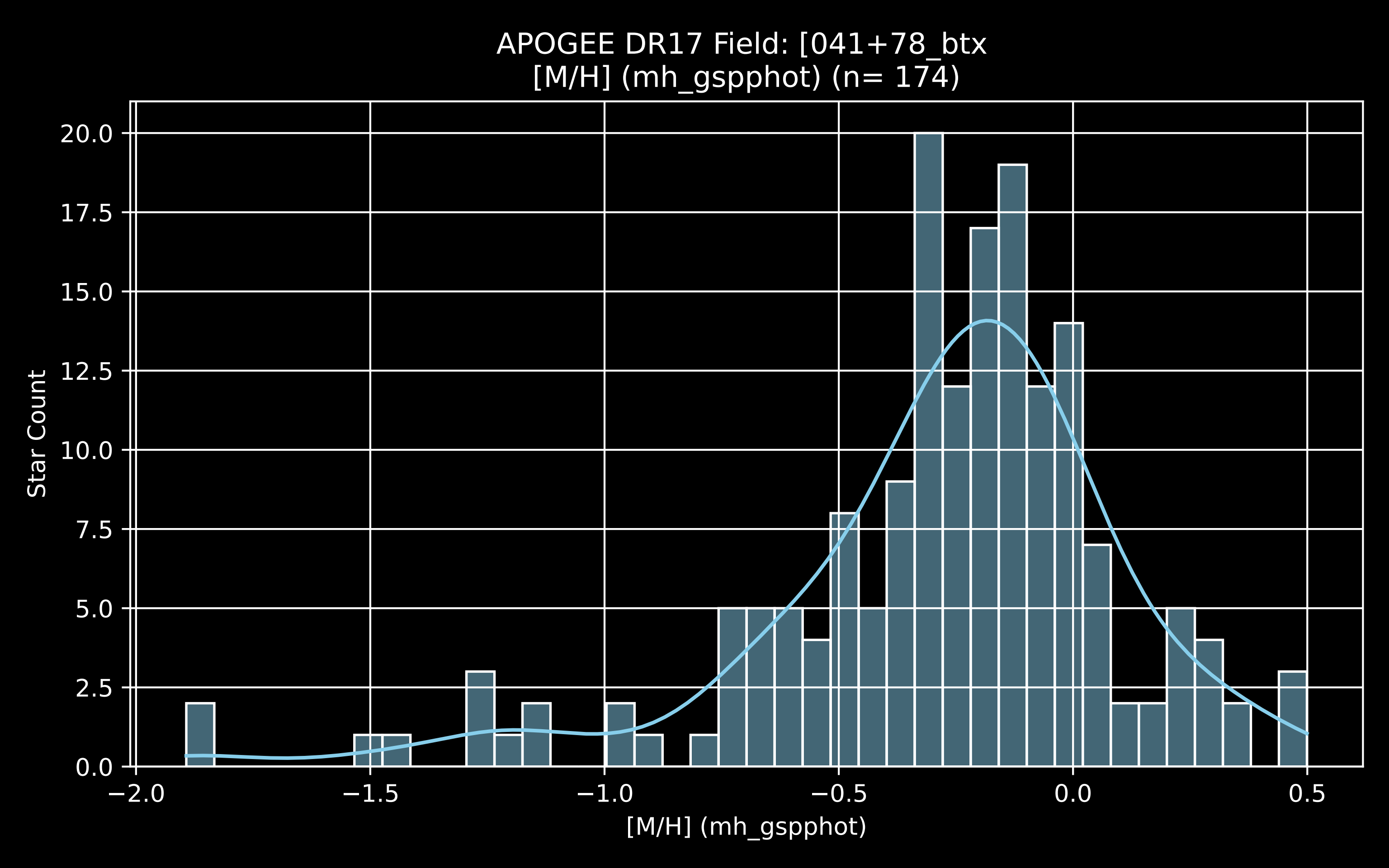

APOGEE DR17 Field [041+78_btx]
Stars (n= 176)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
−4
−2
0
2
4
6
8
10
Color (bp - rp)
0.0
0.5
1.0
1.5
2.0
2.5

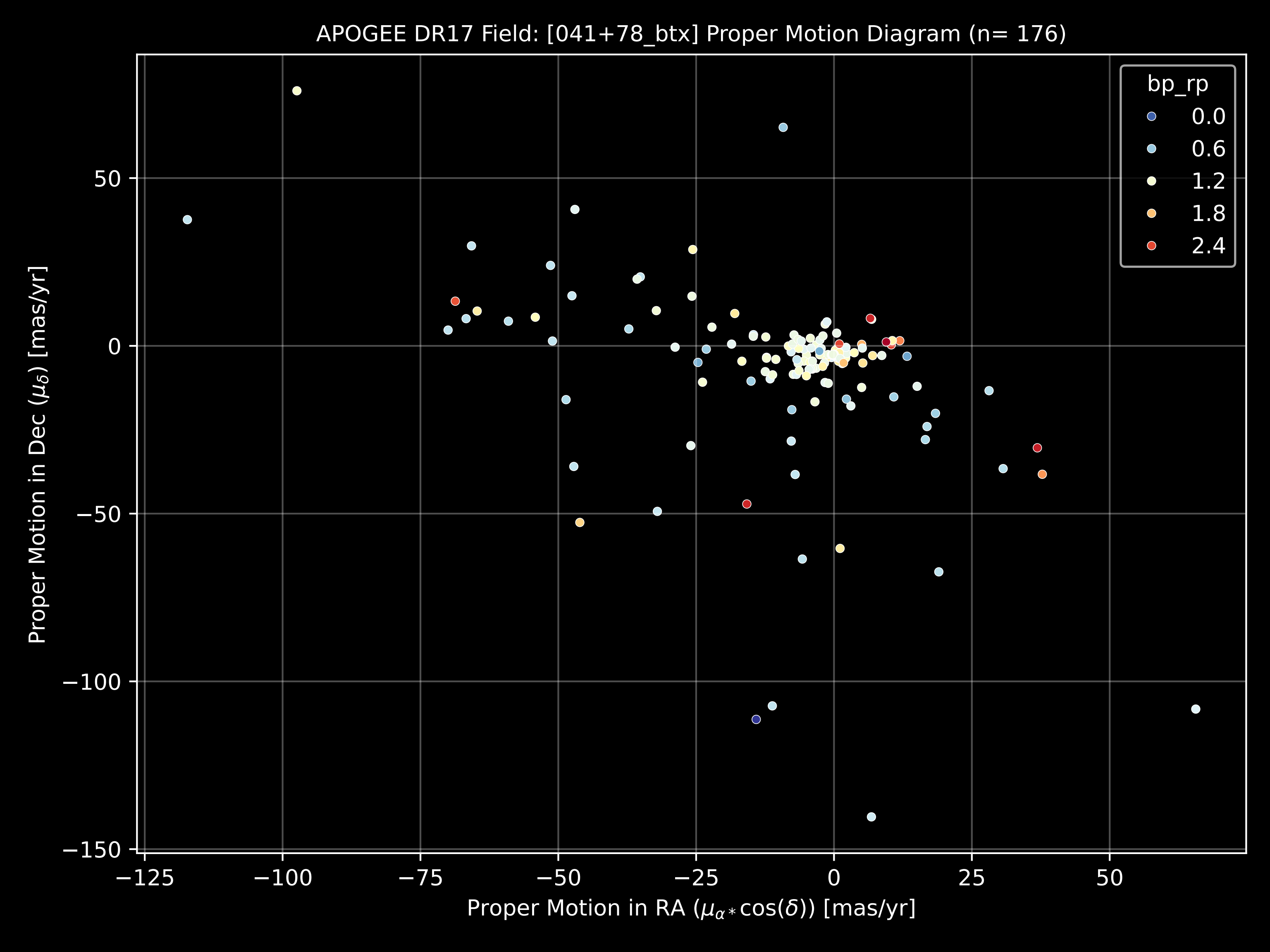

APOGEE DR17 Field: [041+78_btx]
Stars: 176
Dec
Ra
bp_rp
29.5
29.0
28.5
28.0
27.5
27.0
304.0
304.5
305.0
305.5
306.0
306.5
307.0
2.5
2.0
1.5
1.0
0.5
0.0


APOGEE DR17 Field: [041+78_btx]
Stars: 176
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
29.5
29.0
28.5
28.0
27.5
27.0
304.0
304.5
305.0
305.5
306.0
306.5
307.0
2.5
2.0
1.5
1.0
0.5
0.0

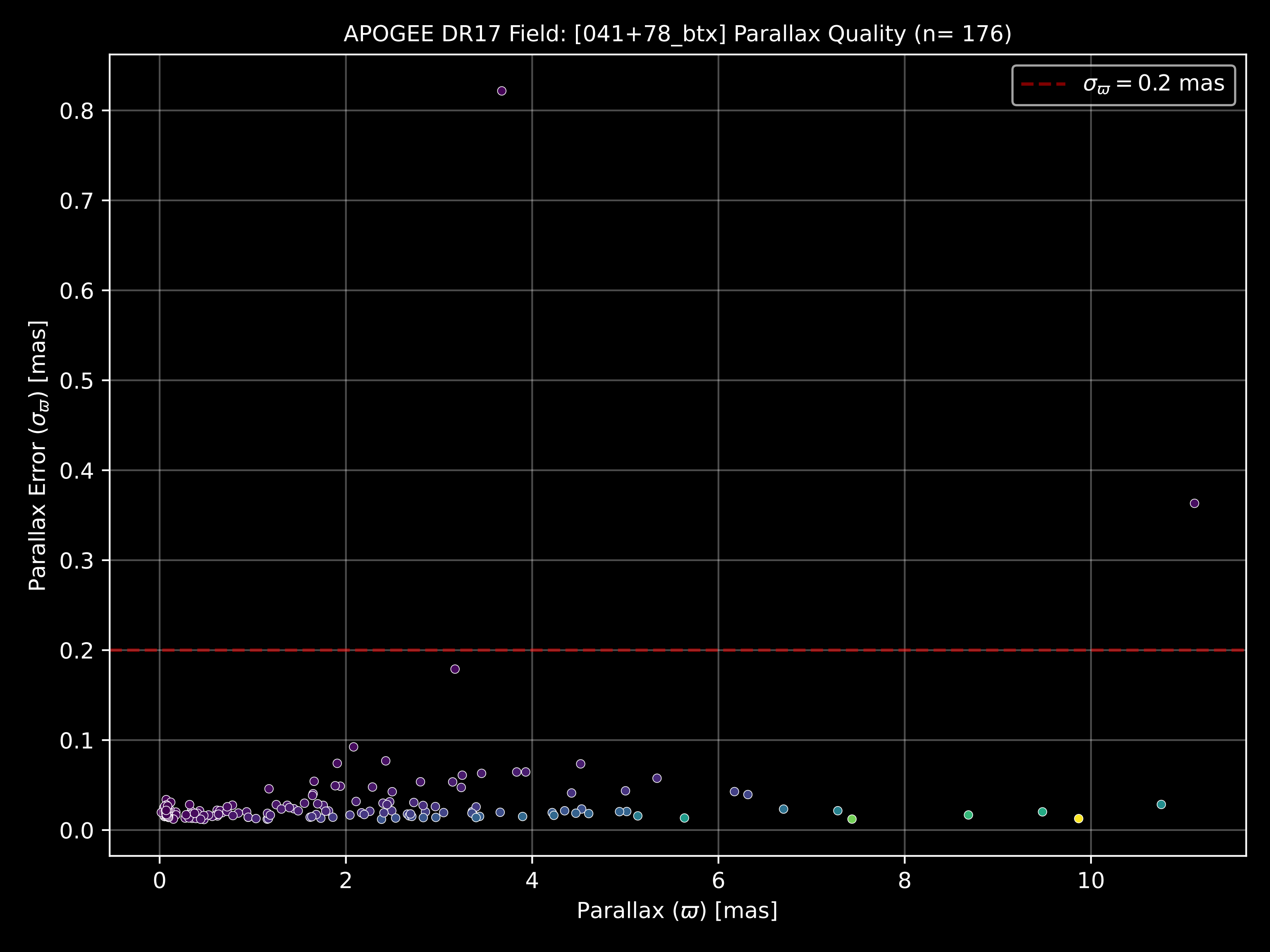

APOGEE DR17 Field: [041+78_btx]
Count plot of Star Type
Count
0
10
20
30
40
50
60
70
80
main_sequence
red_giants
subgiants
turnoff
red_clump
Star Type

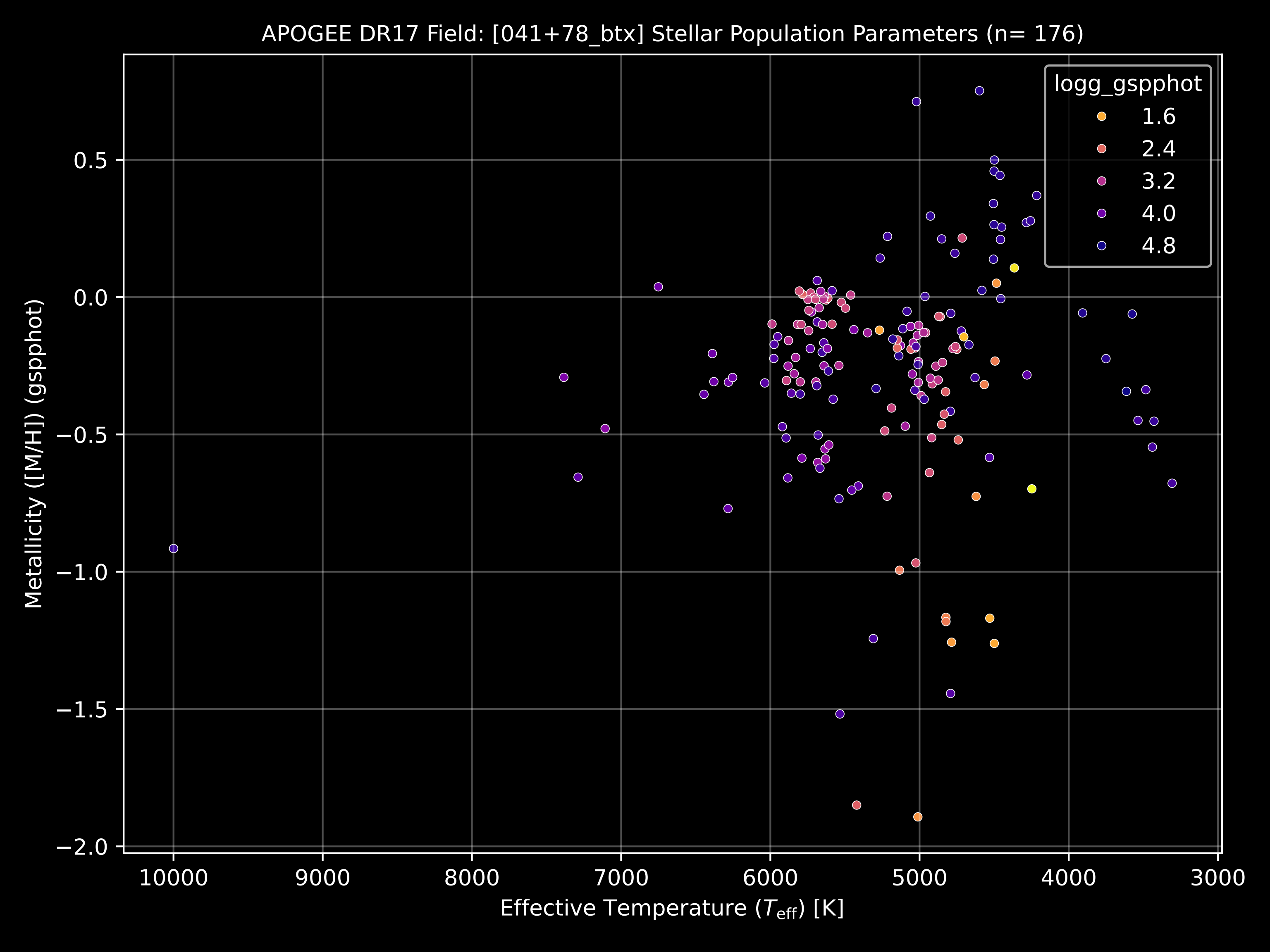


APOGEE DR17 Field: [041+78_btx
[M/H] (mh_gspphot) (n= 174)
Star Count
[M/H] (mh_gspphot)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5

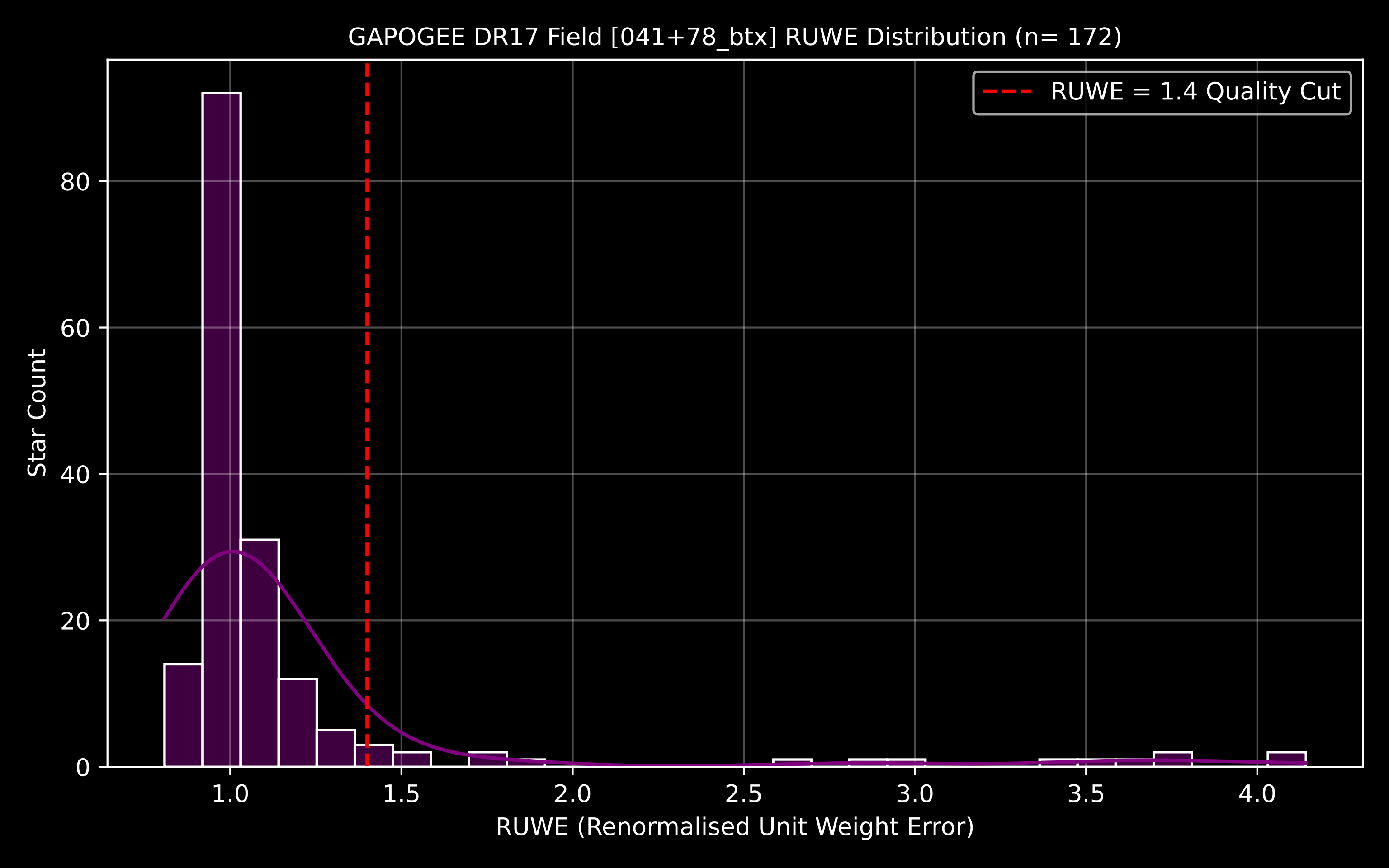


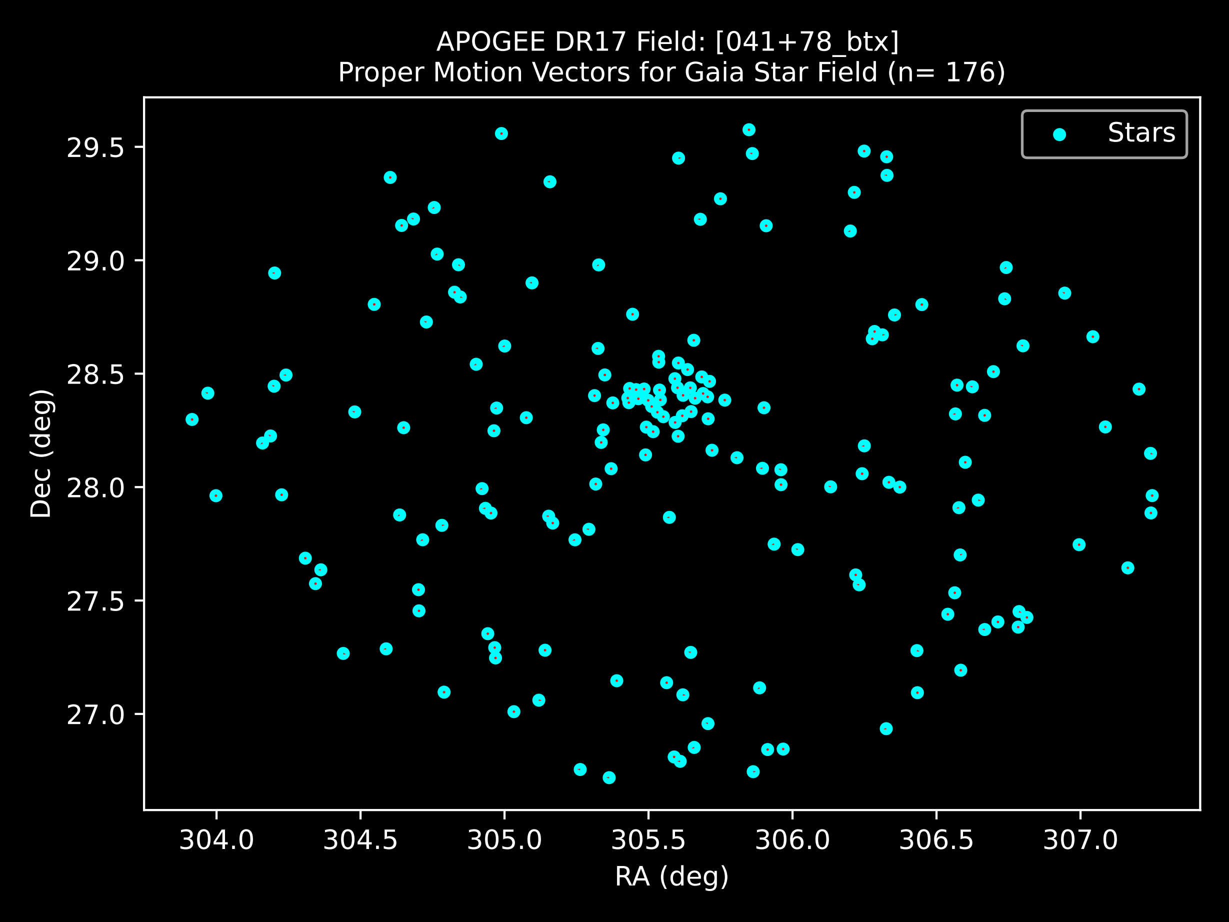
APOGEE DR17 Field: [041+78_btx] Proper Motion Diagram (n= 176)
bp_rp
0.0
0.6
1.2
1.8
2.4
50
0
−50
−100
−150
−125
−100
−75
−50
−25
0
25
50
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]

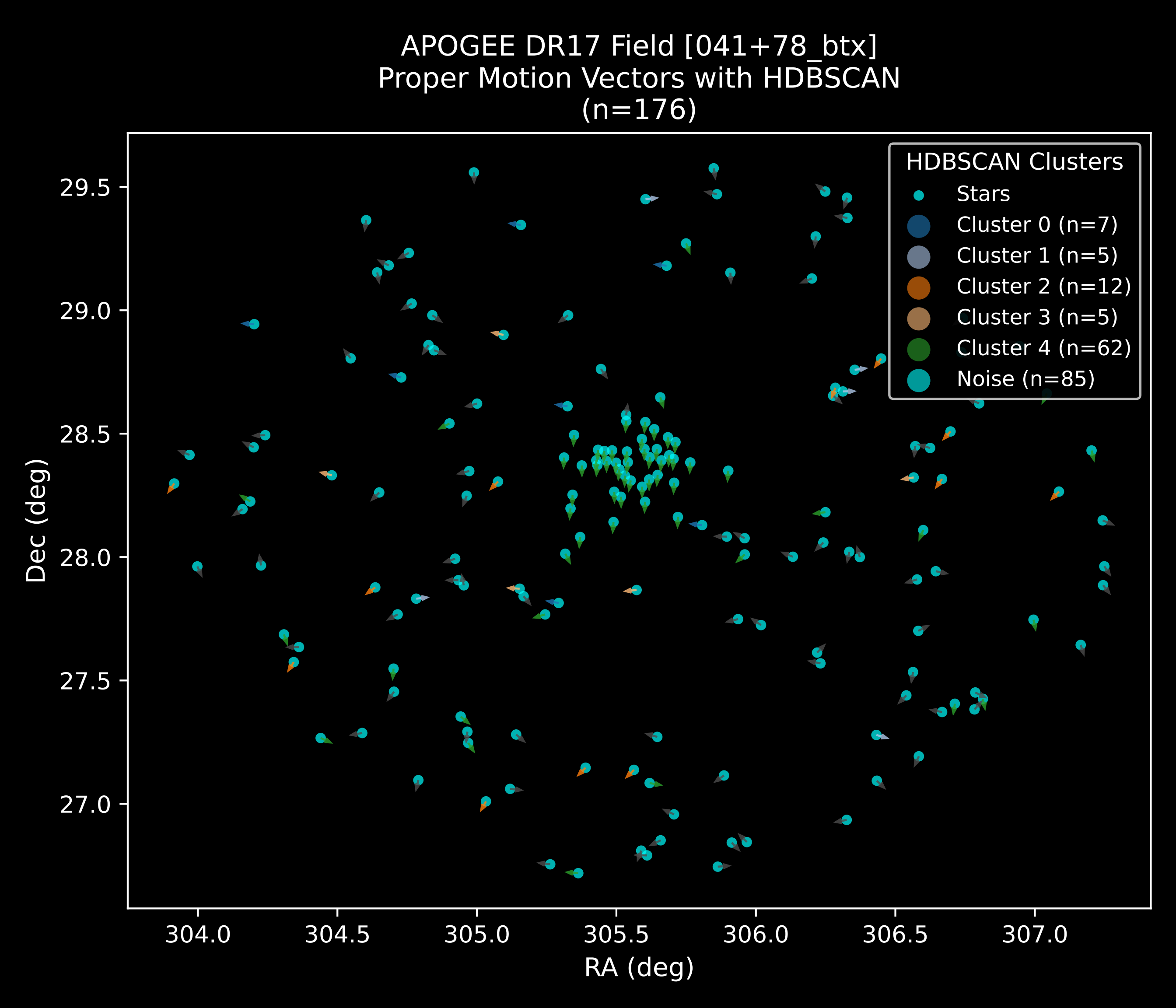

APOGEE DR17 Field: [041+78_btx] Parallax Quality (n= 176)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
2
4
6
8
10

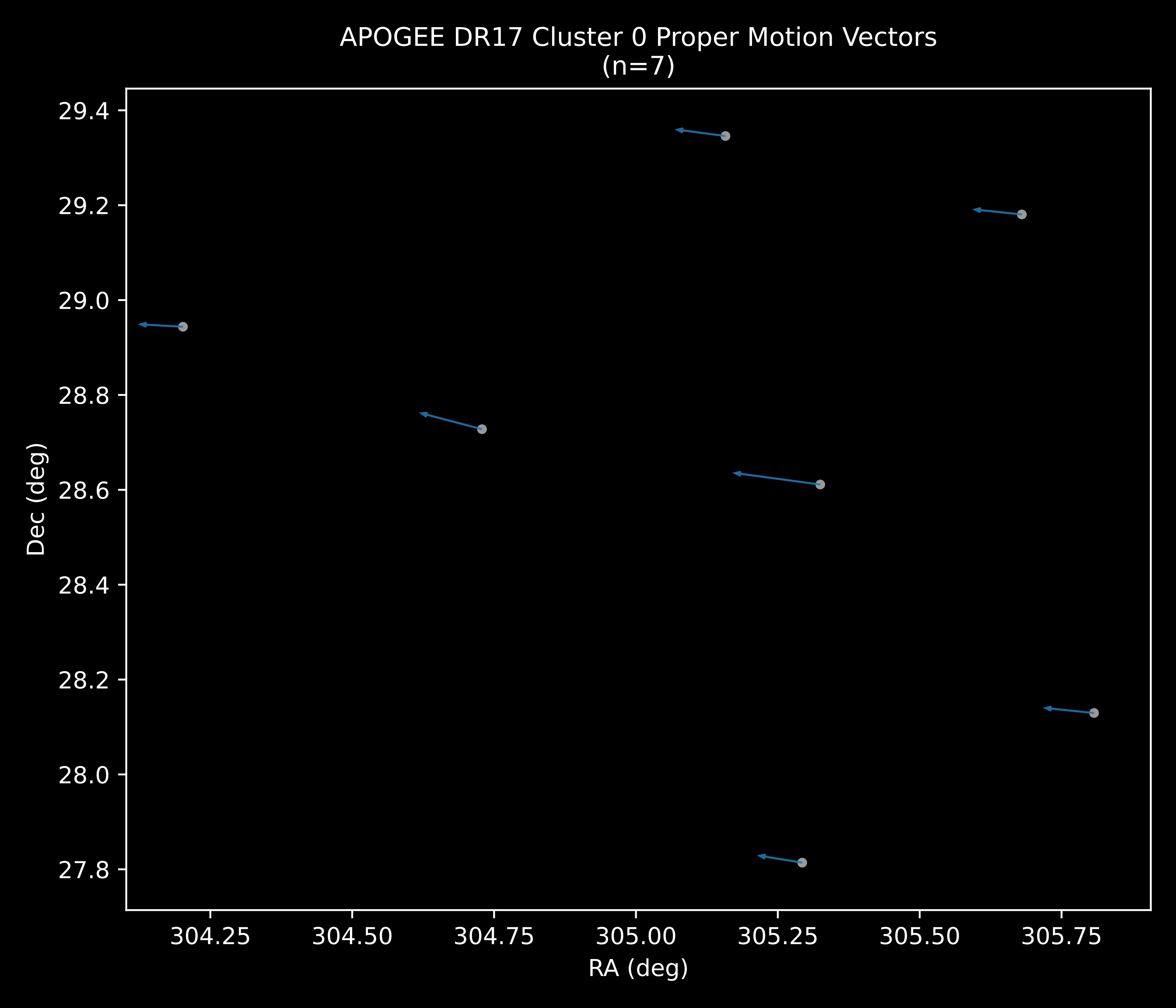


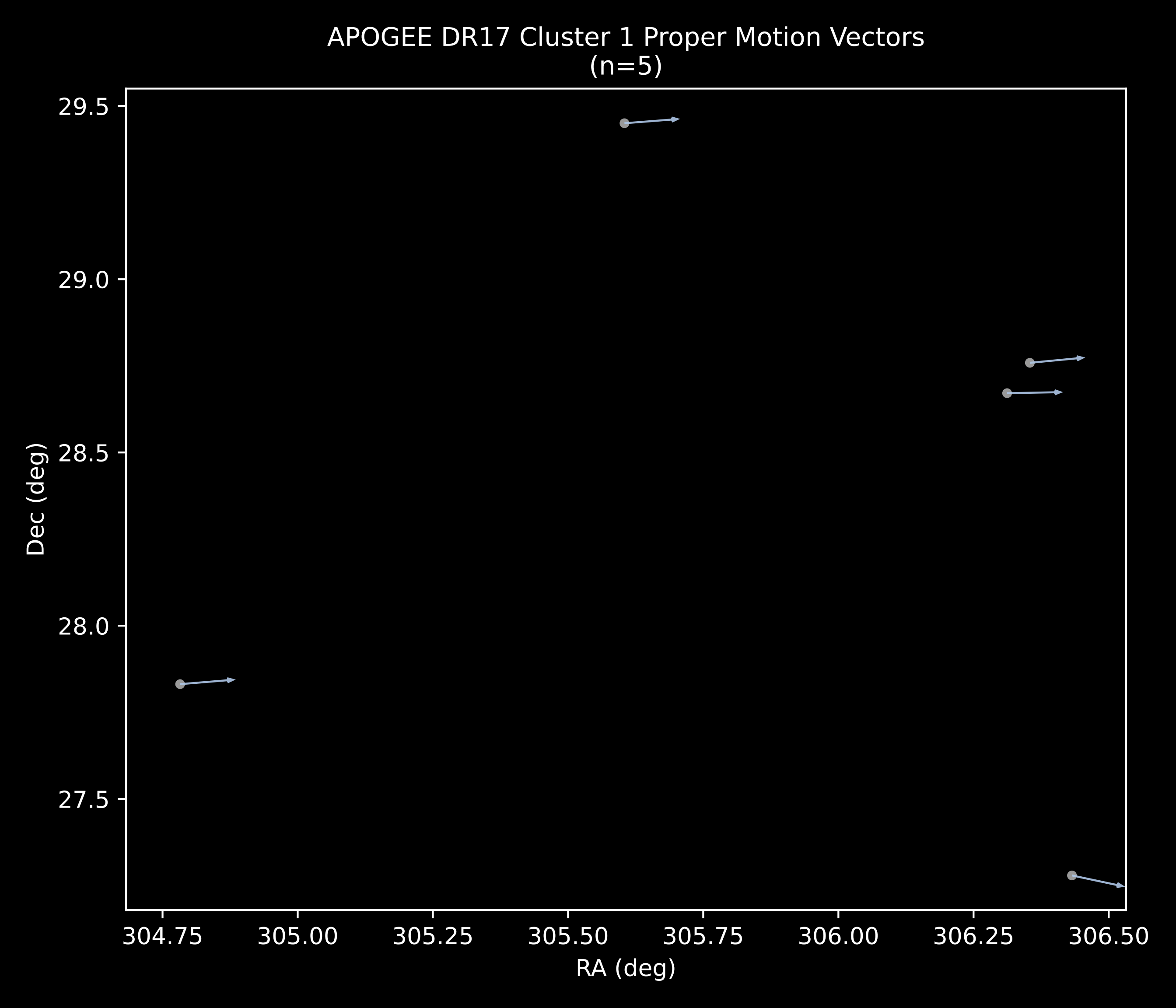
APOGEE DR17 Field: [041+78_btx] Stellar Population Parameters (n= 176)
logg_gspphot
1.6
2.4
3.2
4.0
4.8
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Metallicity ([M/H]) (gspphot)
10000
9000
8000
7000
6000
5000
4000
3000
Effective Temperature (T_{eff}) [K]


GAPOGEE DR17 Field [041+78_btx] RUWE Distribution (n= 172)
RUWE = 1.4 Quality Cut
Star Count
0
20
40
60
80
1.0
1.5
2.0
2.5
3.0
3.5
4.0
RUWE (Renormalised Unit Weight Error)

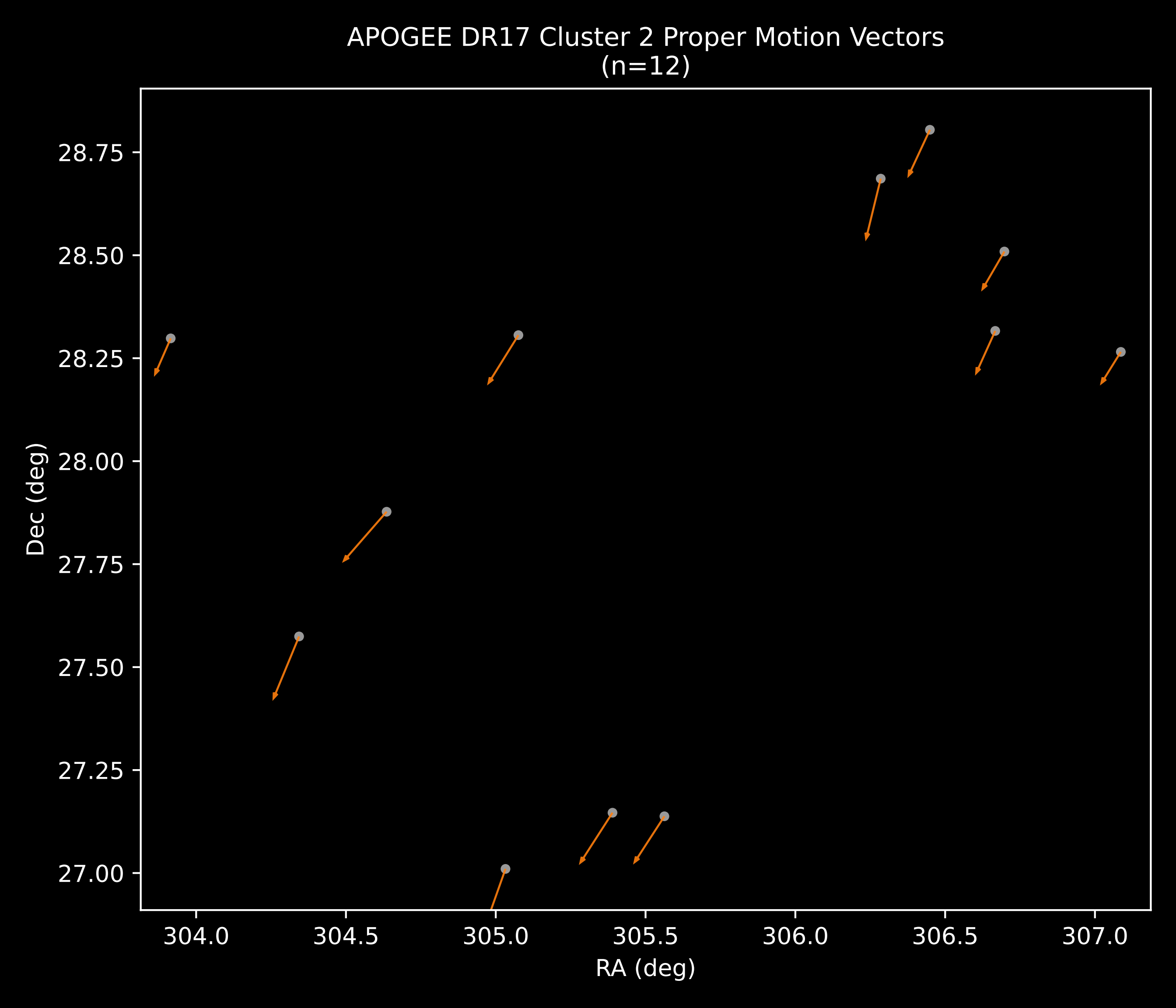

APOGEE DR17 Field: [041+78_btx]
Proper Motion Vectors for Gaia Star Field (n= 176)
Stars
29.5
29.0
28.5
28.0
27.5
27.0
Dec (deg)
304.0
304.5
305.0
305.5
306.0
306.5
307.0
RA (deg)

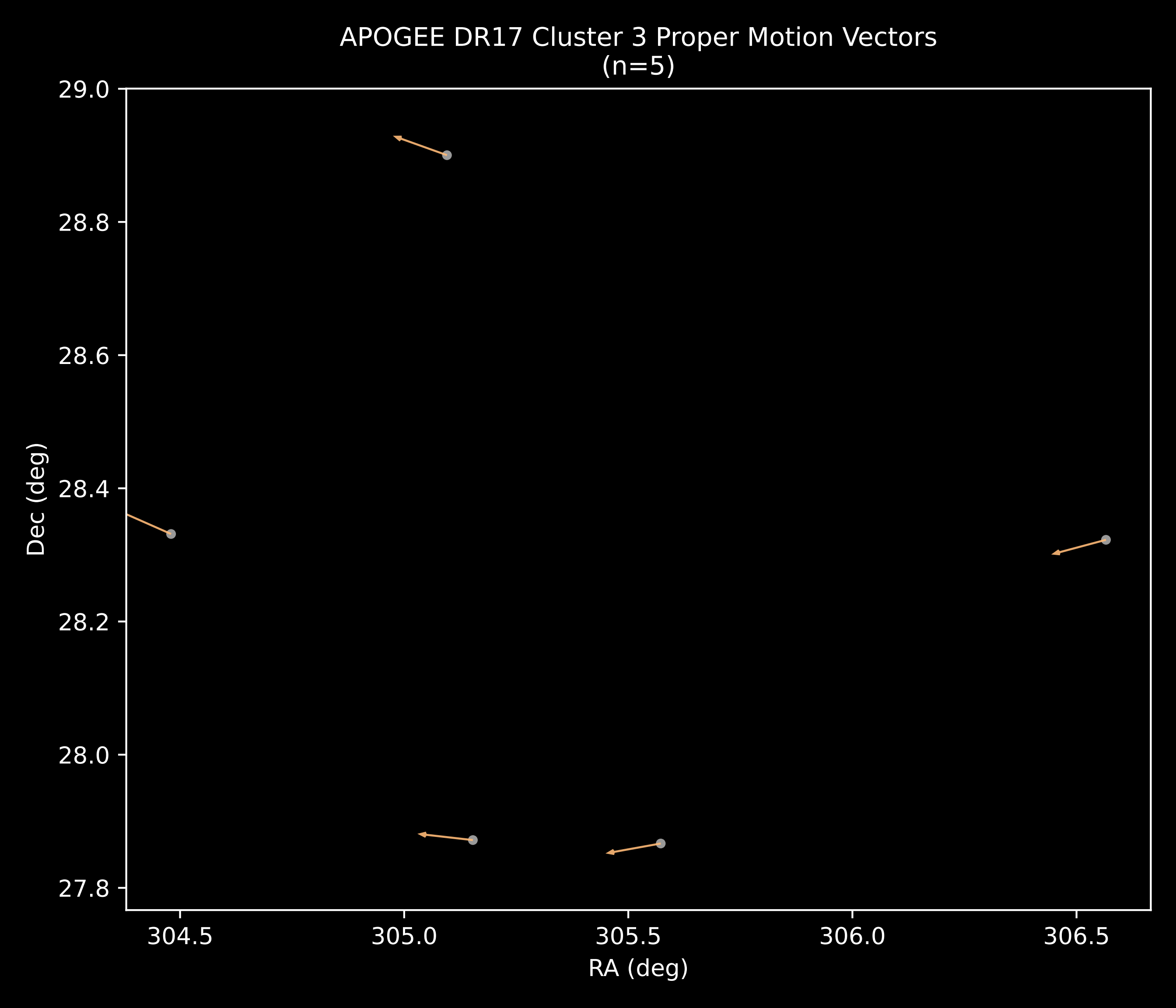

APOGEE DR17 Field [041+78_btx]
Proper Motion Vectors with HDBSCAN
(n=176)
HDBSCAN Clusters
Stars
Cluster 0 (n=7)
Cluster 1 (n=5)
Cluster 2 (n=12)
Cluster 3 (n=5)
Cluster 4 (n=62)
Noise (n=85)
Dec (deg)
RA (deg)
29.5
29.0
28.5
28.0
27.5
27.0
304.0
304.5
305.0
305.5
306.0
306.5
307.0

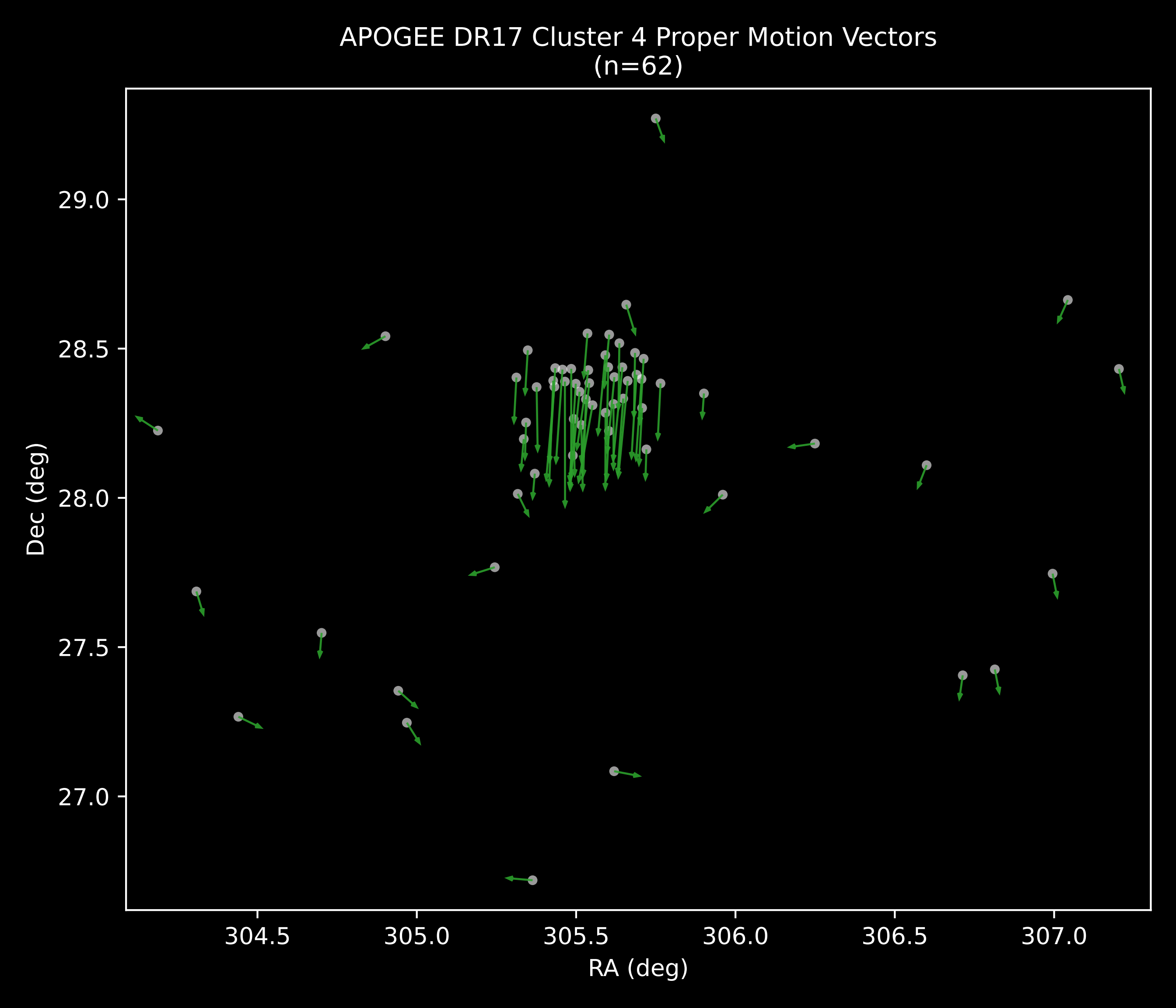

APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=7)
29.4
29.2
29.0
28.8
28.6
28.4
28.2
28.0
27.8
Dec (deg)
304.25
304.50
304.75
305.00
305.25
305.50
305.75
RA (deg)

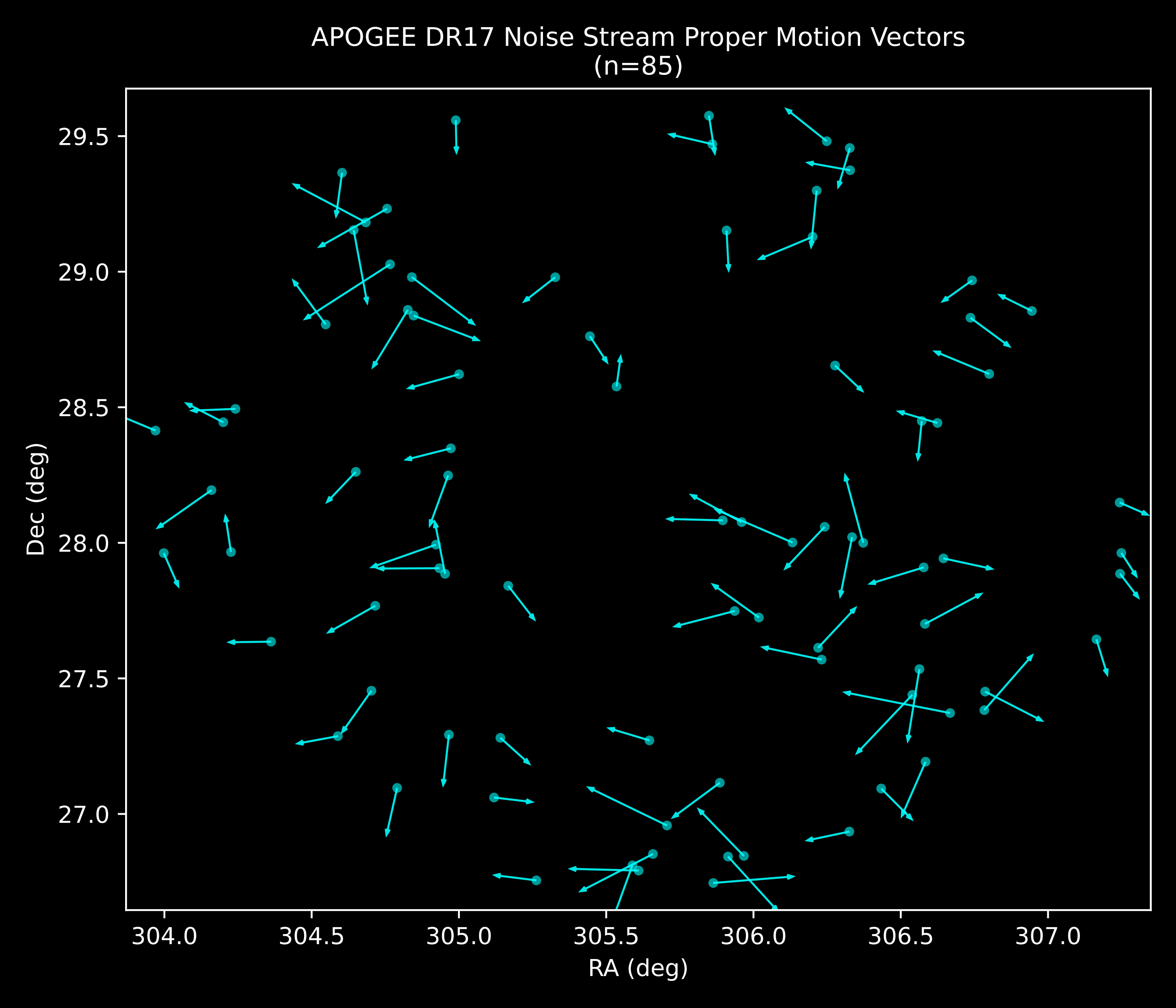


APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=5)
Dec (deg)
RA (deg)
29.5
29.0
28.5
28.0
27.5
304.75
305.00
305.25
305.50
305.75
306.00
306.25
306.50


APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=12)
Dec (deg)
RA (deg)
28.75
28.50
28.25
28.00
27.75
27.50
27.25
27.00
304.0
304.5
305.0
305.5
306.0
306.5
307.0


APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=5)
Dec (deg)
RA (deg)
29.0
28.8
28.6
28.4
28.2
28.0
27.8
304.5
305.0
305.5
306.0
306.5


APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=62)
Dec (deg)
RA (deg)
29.0
28.5
28.0
27.5
27.0
304.5
305.0
305.5
306.0
306.5
307.0


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=85)
Dec (deg)
29.5
29.0
28.5
28.0
27.5
27.0
RA (deg)
304.0
304.5
305.0
305.5
306.0
306.5
307.0